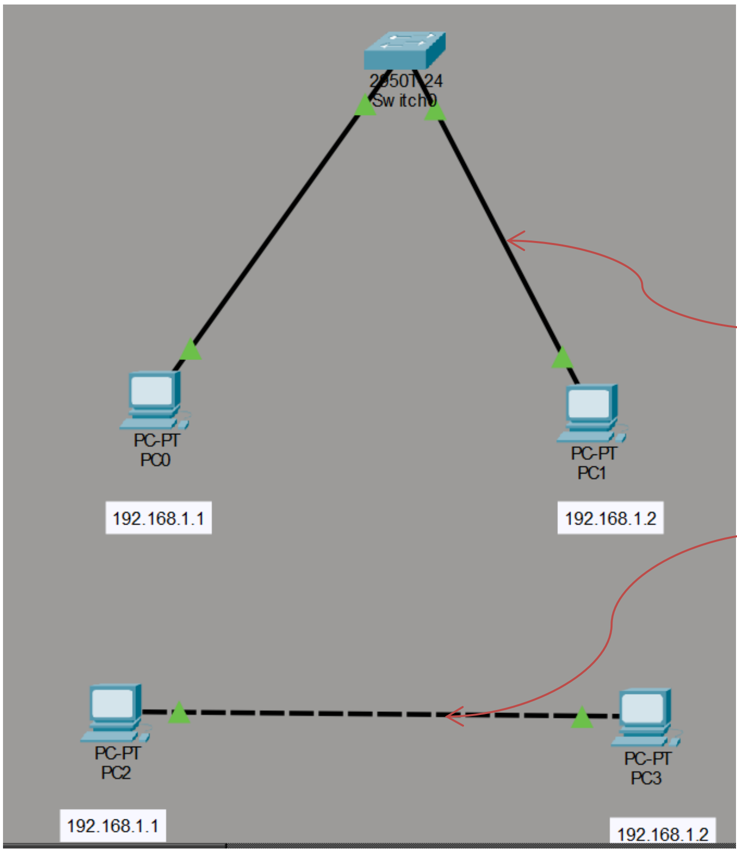
1.Network Topology- PEER TO PEER,SWITCH



. You have to configure both PC's with IP addresses as the image follows,

. Before that make the connection using 1 Switch & 2 or more PCs

. Copper Straight-Through Cable :

To connect different devices

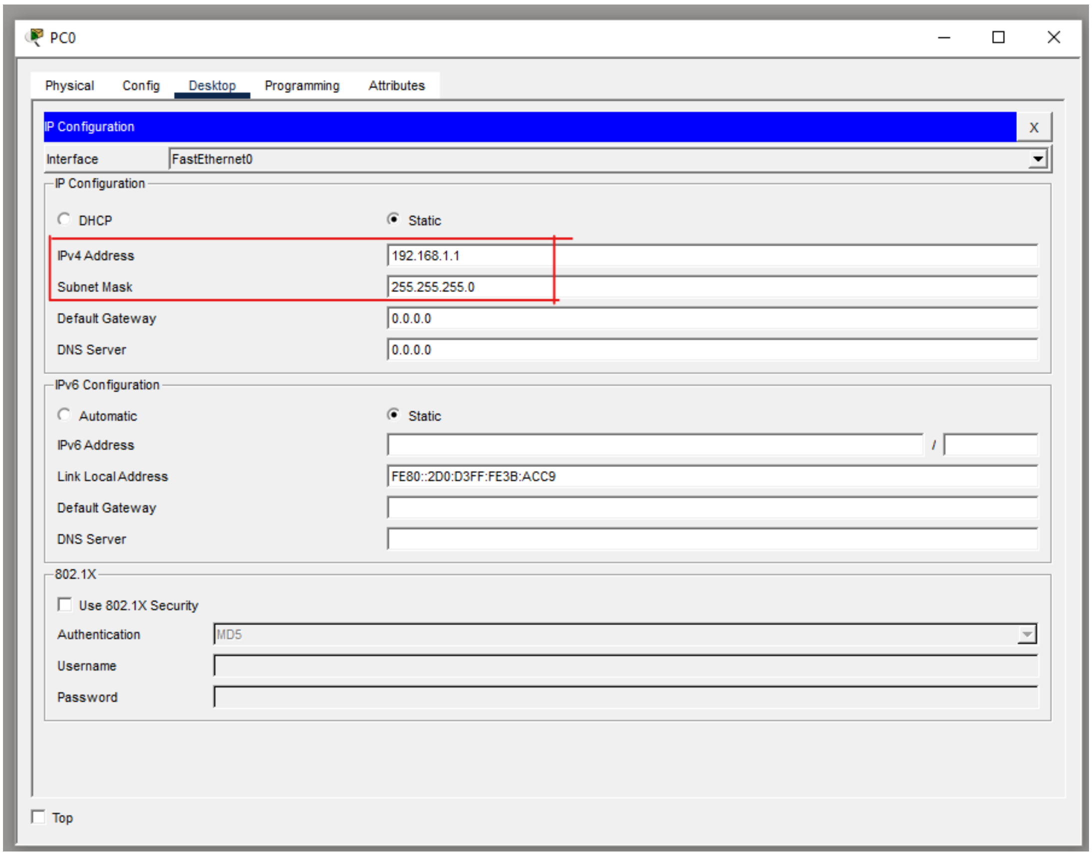
(EX : Switch -> PC)

. Copper Cross-Over Cable:

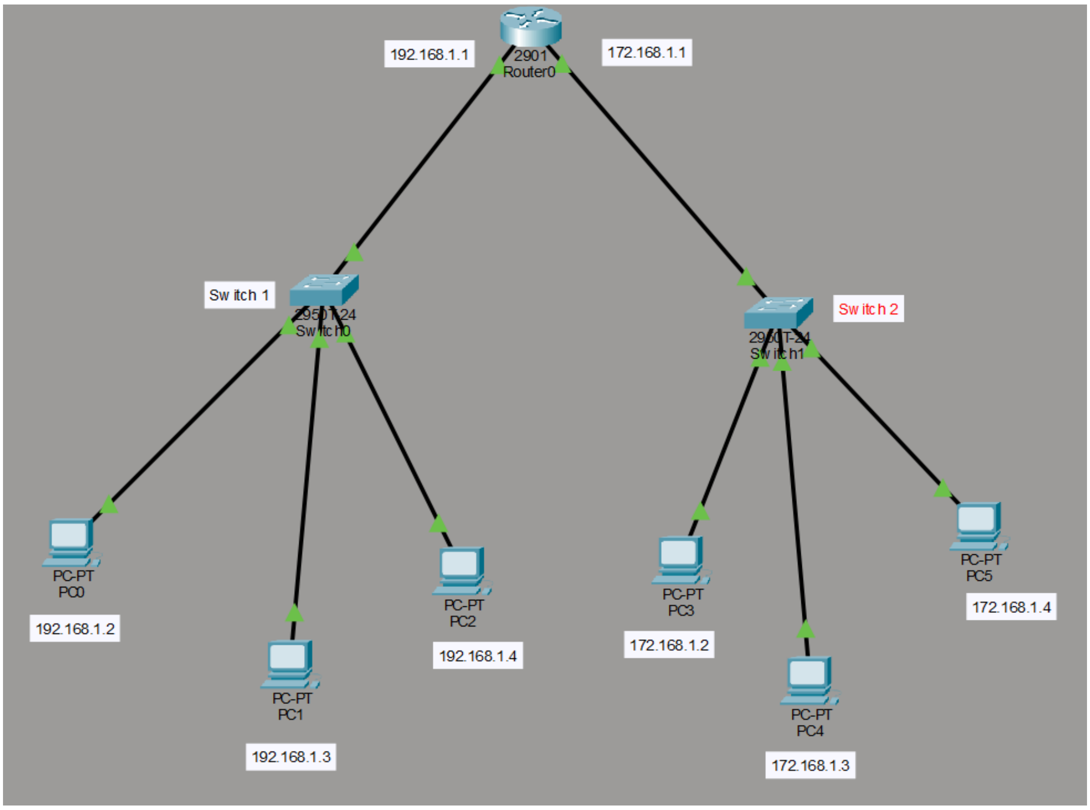
Used to connect same devices

(EX : PC -> PC)

. Note :
Use same class IPs



2.Network Topology – using ROUTER

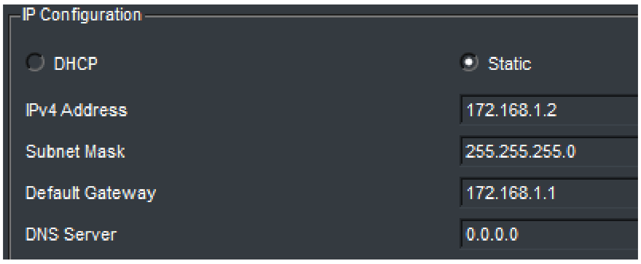


For Switch 1

For Switch 2

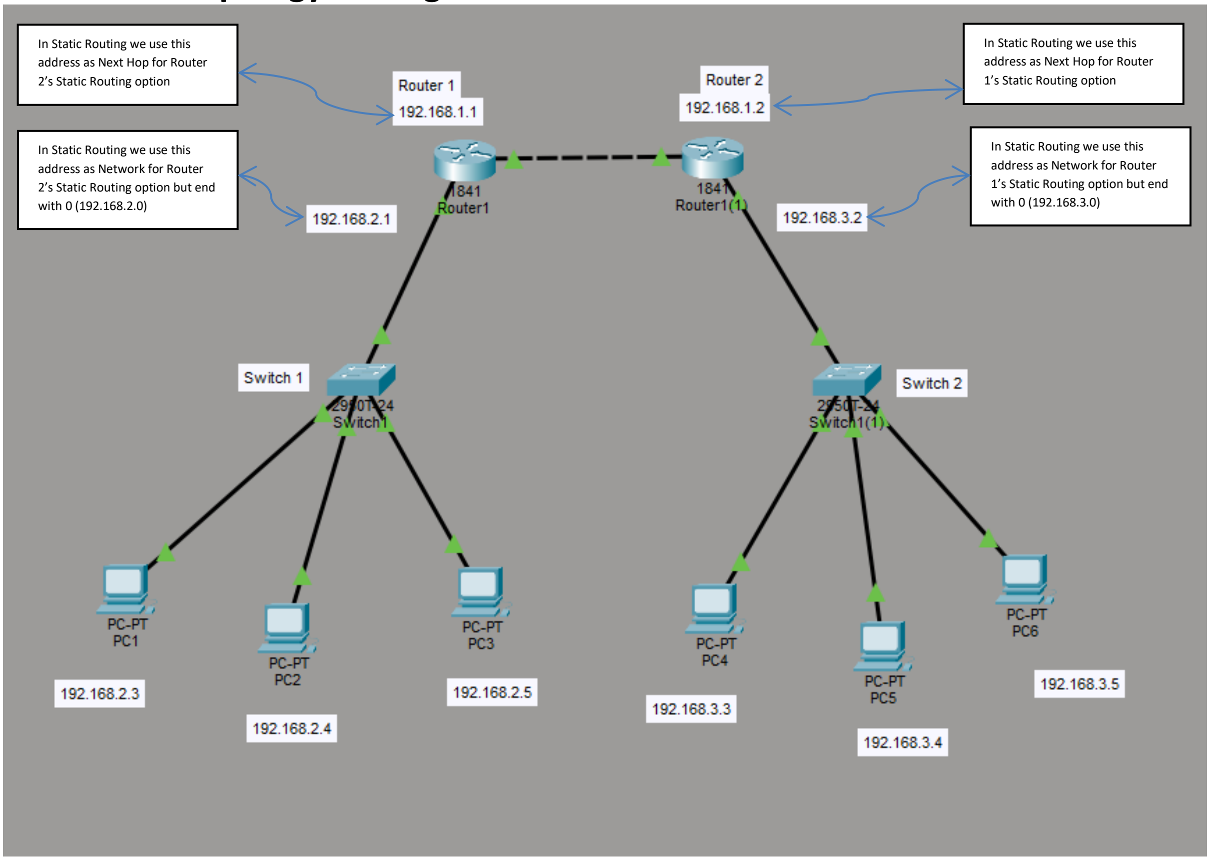
PCs under Switch 1

PCs under Switch 2



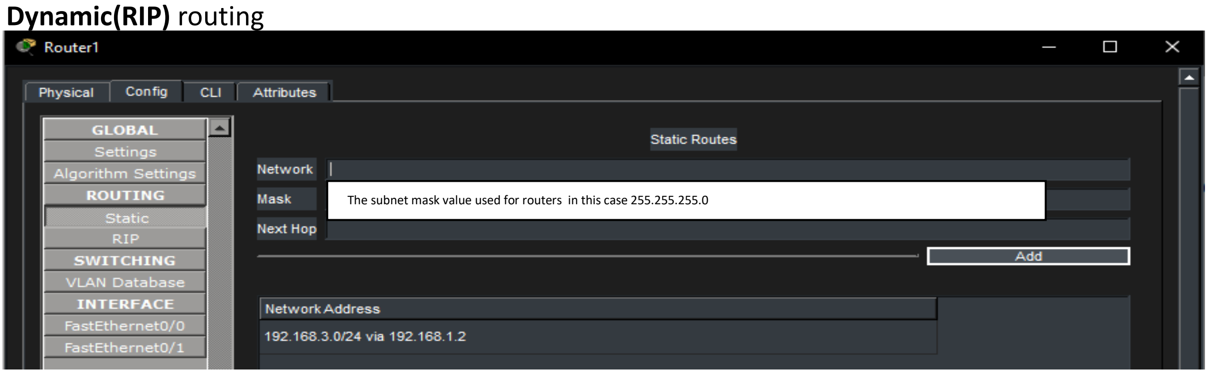
Configure PCs and Router as the image shows.

3.Network Topology – using 2 ROUTER



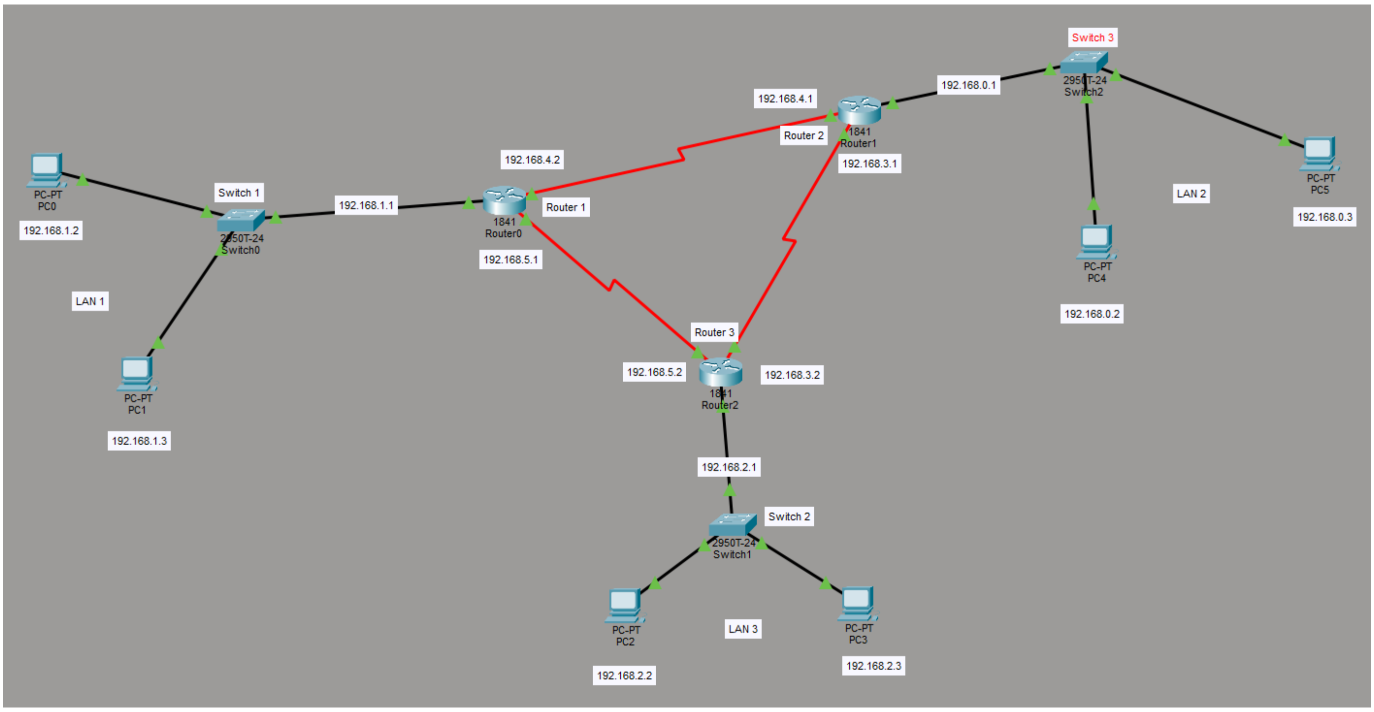
In this case I've used same class IP addresses you can use different IPs

. to make the two Routers communicate with each other, we can use Routing option either **Static** or **Dynamic(RIP)** routing



after all of these routing configure all PCs with corresponding IP addresses, Subnet mask and Default gateway(IP of parent router)

4.Network Topology – using 3 ROUTER



*NOTE:

The red color zig-zag cable used in here is **serial DCE**

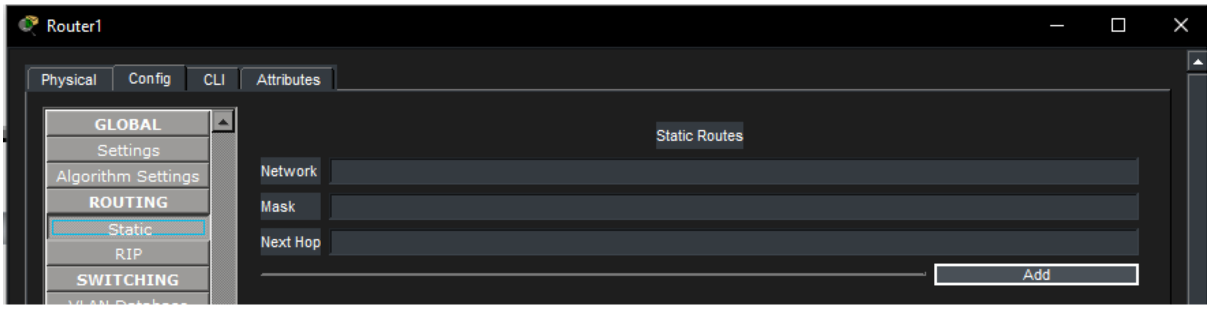
A single Serial cable only has two ends. Therefore, one cable can only connect two specific router interfaces together. In the above figure, you can see this in action:

- ✓ Router0 is connected to Router1.
- ✓ Router1 is connected to Router2.
- ✓ Router2 is connected to Router0.

Together, they form a triangle, but each "side" of that triangle is a separate, individual serial link requiring its own subnet and pair of interfaces.

To make this connection, we can use the RIP(Dynamic) or the Static method of routing

You have to configure each routers separately by clicking on router go to config menu then the static tab under routing option,



On each routers make sure to add 2 addresses (not only one as we did in previous experiment)

Router 1 need to communicate with router 2 & 3 and all others have to communicate like this so->

According to the topology shown above :

<u>Router 1</u>

Network = Network of Router 2 (192.168.0.0)

Mask = 255.255.255.0

Next Hop = IP of Router 2 (192.168.4.1), (same end of DCE cable associating with router 1)

Click on the Add button then create another table as follows

Network = Network of Router 3 (192.168.2.0)

Mask = 255.255.255.0

Next Hop = IP of Router 3 (192.168.5.2), (same end of DCE cable associating with router 1)

<u>Router 2</u>

Network = Network of Router 1 (192.168.1.0)

Mask = 255.255.255.0

Next Hop = IP of Router 1 (192.168.4.2), (same end of DCE cable associating with router 2)

Click on the Add button then create another table as follows

Network = Network of Router 3 (192.168.2.0)

Mask = 255.255.255.0

Next Hop = IP of Router 3 (192.168.3.2), (same end of DCE cable associating with router 2)

<u>Router 3</u>

Network = Network of Router 1 (192.168.1.0)

Mask = 255.255.255.0

Next Hop = IP of Router 1 (192.168.5.1), (same end of DCE cable associating with router 2)

Click on the Add button then create another table as follows

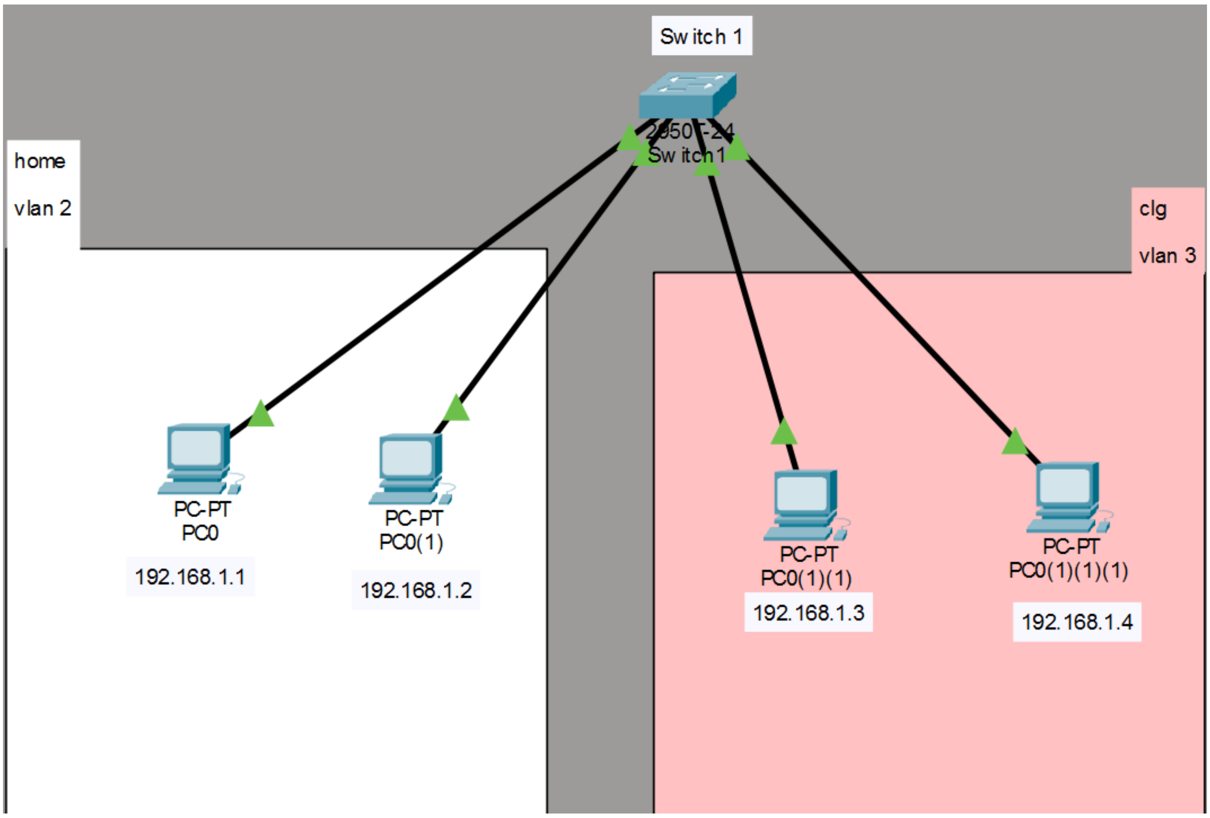
Network = Network of Router 2 (192.168.0.0)

Mask = 255.255.255.0

Next Hop = IP of Router 2 (192.168.3.1), (same end of DCE cable associating with router 2)

Okay that's it the routing has done successfully

5.Network Topology – VLAN config



First make a simple connection and open the CLI of switch and enter following commands as follows

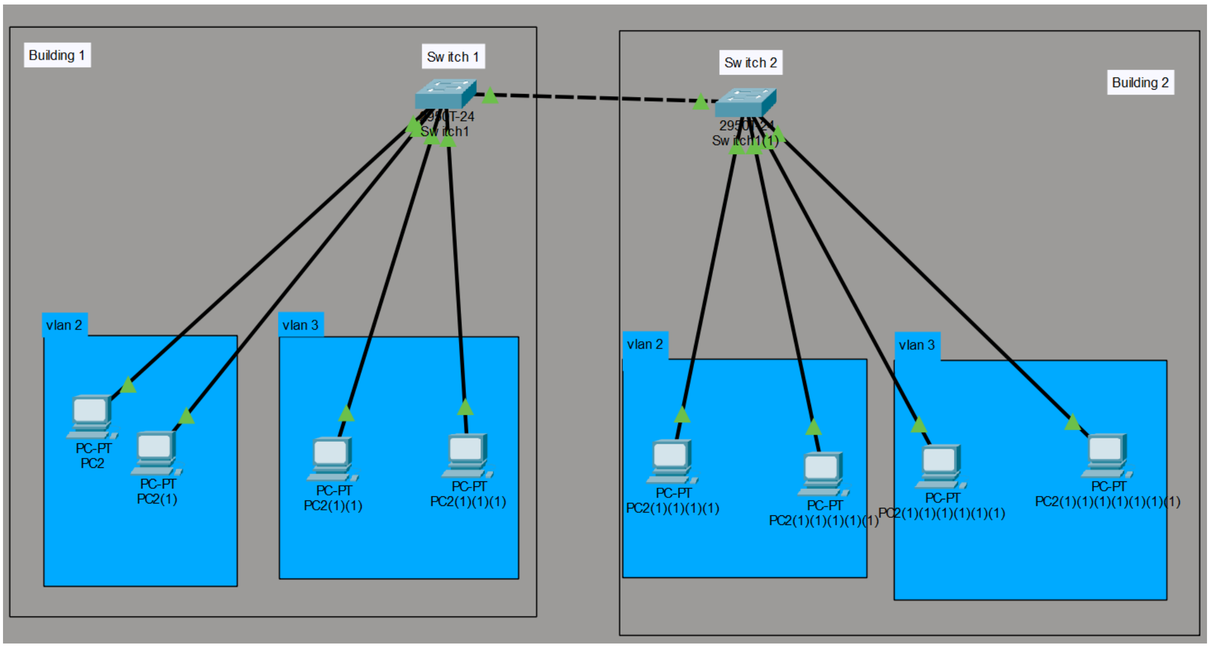
1. en	4. name home	7. Name clg	10. Sw ac vlan 2	13. Sw ac vlan 3
2. conf t	5. Exit	8. Exit	11. Exit	14. exit
3. vlan 2	6. Vlan 3	9. Int range fa0/1-2	12. Int range fa0/3-4	15. End

*NOTE

To make the command work you have to make the exact connection as in the imageand connecting the cables from switch to PCs have to be **in order** like switch->PC1, switch->PC2, switch->PC3, Switch->PC4.

And configure the PCs with appropriate IP addresses

6. Network Topology - VLAN using TRUNK



First make a simple connection and open the CLI of switch 1 and enter following commands as follows:

1. en	5. exit	9.int range fa0/1-2	13. int range fa0/3-4	17. exit
2. conf t	6. vlan 3	10. sw mo acc	14. sw mo acc	18. int fa0/5
3. vlan 2	7. name lab	11. sw acc vlan 2	15. sw acc vlan 3	19. sw mo trunk
4. name Home	8. exit	12. exit	16. Exit	

Then open CLI of Switch 2 and follow the commands:

1. en	5. exit	9.int range fa0/1-2	13. int range fa0/3-4	17. exit
2. conf t	6. vlan 3	10. sw mo acc	14. sw mo acc	18. int fa0/5
3. vlan 2	7. name lab	11. sw acc vlan 2	15. sw acc vlan 3	19. sw mo trunk
4. name Home	8. exit	12. exit	16. Exit	

*NOTE

en = enable

conf t = configure terminal

sw mo acc = switchport mode access

sw acc = switchport access

int fa0/5 = interface FastEthernet0/5

also there is GigabitEthernet = int gi0/1